Víctor Humberto Casanova Bernal

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Final-year Industrial Electronics and Automation Engineering student (UPM) with a hands-on hardware and software profile: robotics, microcontrollers and embedded systems, with training in FPGA design (VHDL). Currently an intern at NTT DATA (cloud and DevOps). Interested in embedded systems, FPGA and robotics development.

EXPERIENCE

NTT DATA · AI Engineering Talent Factory — **Madrid, Spain**
Digital Architect (Intern) — *Oct 2026 – Present*

- Infrastructure as code on Microsoft Azure with Terraform, and CI/CD pipelines with GitHub Actions.
- Containers with Docker and Kubernetes; AI-assisted development (Claude Code) with Git and code review.

CREA-ETSIDI (UPM Robotics Club) — **Madrid, Spain**
Robotics Developer — *Feb 2023 – Present*

- Instructor of the club's official Arduino course: electronics and C++ programming.
- Design, assembly and programming of robots, from RC vehicles to TunaBot (with CSIC) and EUROBOT 2027.

UPM MotoStudent Petrol — **Madrid, Spain**
Electronics Department Member — *Nov 2024 – Jun 2026*

- Development of a custom telemetry dashboard that reads the motorbike's OBD-II bus.
- Schematics and documentation in OrCAD; diagnosis and repair of electronic faults during competition.

FEATURED PROJECTS

TunaBot: swarm of coordinated robots (CSIC) — **Madrid, Spain**
Locomotion Team · CREA-ETSIDI — *Apr 2026 – Present*

- ESP32 locomotion firmware (C++, PlatformIO) for omnidirectional wheels with vector-based control.
- Robot board design (ESP32, ULN2003 and stepper motors) and debugging of a short circuit.

Computer vision guided robotic arm — **Madrid, Spain**
Hardware and Software Developer — *Feb 2025 – May 2025*

- Mechanical design in Fusion 360 and Arduino control of servos and stepper motors from a PC.
- Real-time vision algorithm (Python, OpenCV) that detects targets and computes their coordinates.
- v2 in progress: headless Linux on a Raspberry Pi 4, PWM over I2C (PCA9685) and video streaming.

Digital system on FPGA and STM32 (in progress) — **Madrid, Spain**
Group project · UPM — *Nov 2026 – Jan 2027*

- Digital system in VHDL: design, testbench simulation and testing on a Nexys 4 DDR FPGA (Vivado).
- C firmware for an STM32F4 (ARM Cortex-M4, HAL); delivery and public presentation in January 2027.

EDUCATION

UNIVERSIDAD POLITÉCNICA DE MADRID (UPM) - ETSIDI — **Madrid, Spain**
BSc in Industrial Electronics and Automation Engineering — *Sep 2022 – Jun 2027*

- **Relevant courses:** Digital Electronic Systems (VHDL and FPGA; C on STM32F4), Control Engineering, Industrial Computing (C++).

AWARDS

- **First National Prize, ALIA 40B Promptathon** (Spanish Ministry for Digital Transformation) · Dec 2025

SKILLS

- **Embedded and FPGA:** C | C++ | ESP32 | STM32 (HAL) | Raspberry Pi | Arduino | VHDL | FPGA (Vivado) | I2C
- **Electronics:** Circuit design | OrCAD | OBD-II bus | Fusion 360
- **Software:** Python | MATLAB | Git/GitHub | Linux | Docker | GitHub Actions | Microsoft Azure | Terraform
- **Languages:** English (B2 level) | Spanish (native)